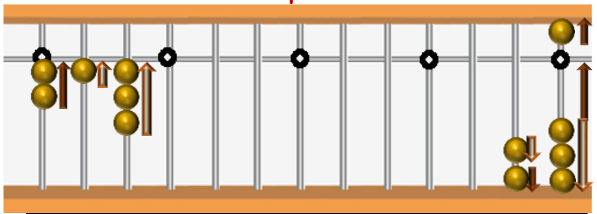
CHAPTER - 2

PRIME FACTORISATION, HCF & LCM

PRIME FACTORISATION

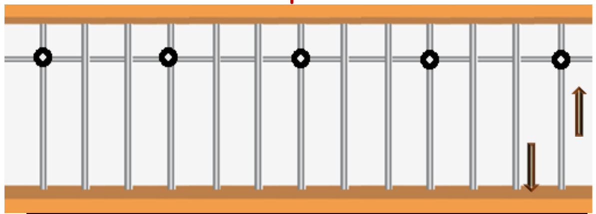
Example: 26



											2	6
2											1	3
	1	3										1

Result : 2,13

Example: 72



											7	2
2											3	6
	2										1	8
		2										9
			3									3
				3								1

Result : 2, 2, 2, 3, 3

PRACTICE SHEET – 2.1

Solve the following and find prime factors:

a)	45	=		e)	24	=		i)	176	=	
b)	63	=		f)	39	=		j)	480	=	
c)	81	=		g)	169	=		k)	540	=	
d)	64	=		h)	144	=		l)	369	=	

DATE: / /

PRACTICE SHEET – 2.2

Solve the following and find prime factors:

a) 333	b) 18
Result :	Result :

c) 69	d) 126
Result :	Result :

e) 152

Result :

f) 1440

Result :

g) 720

Result :

h) 1260

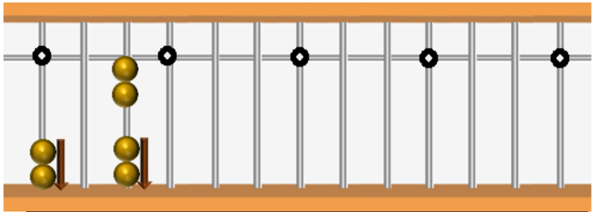
Result :

DATE: / /

HIGHEST COMMON FACTOR [HCF]

Examples:

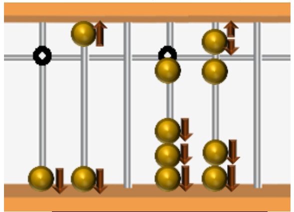
HCF (2,4)



2		4										
2		2										
0		2										

Result : 2

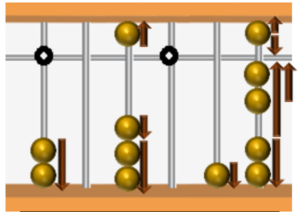
HCF (16,48)



1	6		4	8
1	6		3	2
1	6		1	6
0			1	6

Result :16

HCF (2,8,10)



2		8		1	0
2		6		2	
2		4		2	
2		2		2	
0		0		2	

Result : 2

PRACTICE SHEET – 2.3

Solve the following:

a)	HCF (5 , 15)	=	
b)	HCF (8 , 32)	=	
c)	HCF (16 , 80)	=	
d)	HCF (11 , 33)	=	

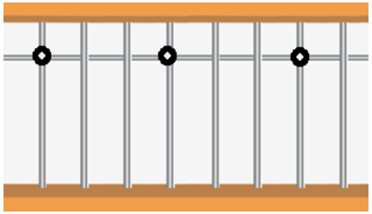
e)	HCF (3 , 6 , 9)	=	
f)	HCF (8 , 16 , 32)	=	
g)	HCF (11 , 33 , 55)	=	
h)	HCF (38 , 19 , 114)	=	

i)	HCF (24 , 36 , 48)	=	
j)	HCF (6 , 15 , 35)	=	
k)	HCF (16 , 24 , 36)	=	
l)	HCF (18 , 45 , 54)	=	

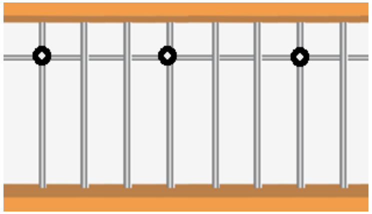
PRACTICE SHEET – 2.4

Solve the following using abacus representation:

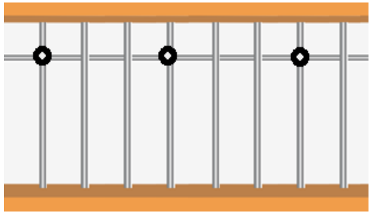
a) HCF (3 , 9 , 12)



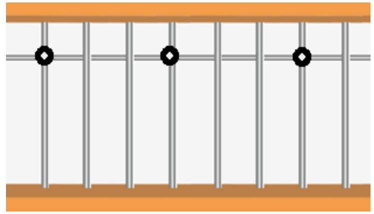
b) HCF (15 , 45)



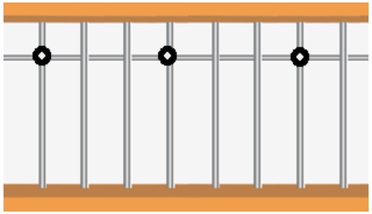
c) HCF (39 , 195)



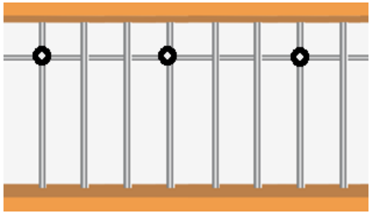
d) HCF (4 , 12 , 18)



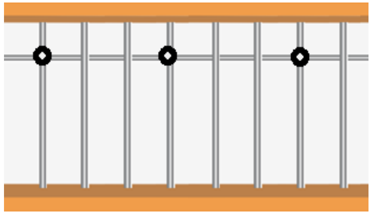
e) HCF (13 , 104)



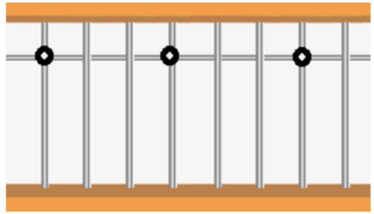
f) HCF (16 , 48 , 72)



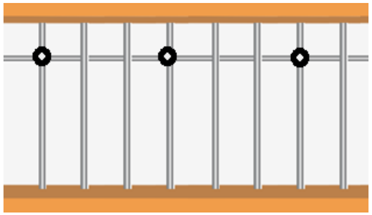
g) HCF (28 , 70)



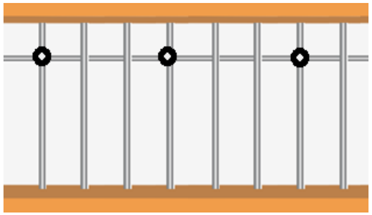
h) HCF (32 , 112)



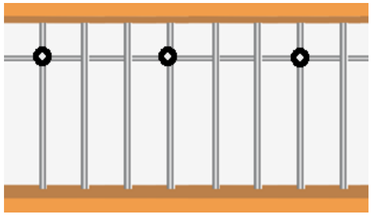
i) HCF (36 , 90)



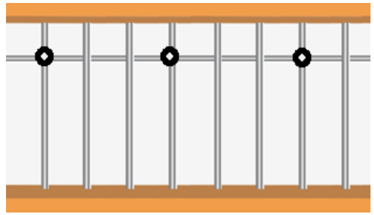
j) HCF (55 , 70 , 60)



k) HCF (48 , 84)



l) HCF (15 , 120)



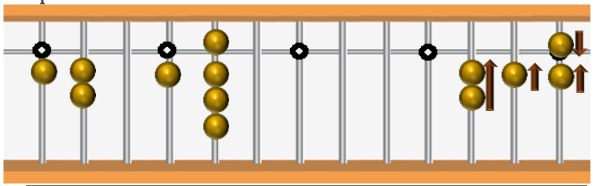
LEAST COMMON MULTIPLE [LCM]

Examples:

METHOD – 1

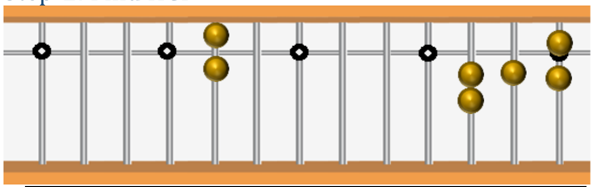
LCM (12,18)

Step-1: Find Product



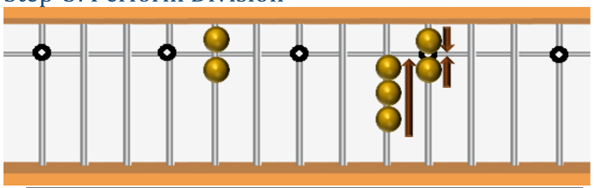
1	2		1	8						2	1	6

Step-2: Find HCF



0	0		0	6						2	1	6

Step-3: Perform Division



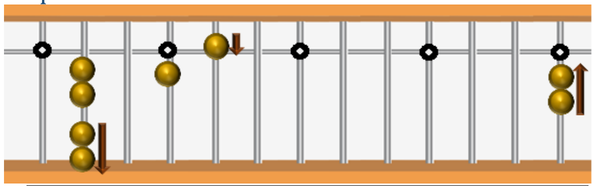
			0	6				3	6	0	0	0

Result : 36

METHOD - 2

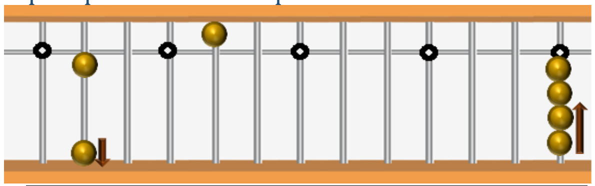
LCM (4,10)

Step-1: Find Prime Factors and divide the numbers

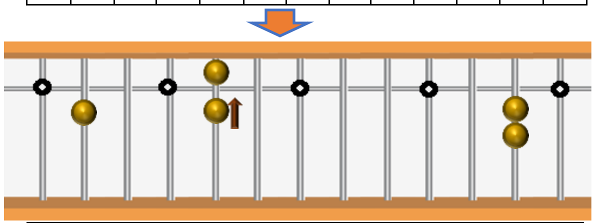


0	4		1	0								
0	2		0	5							0	2

Step-2: Replace the prime factor with latest product and repeat process until 1's present on left side.



0	1		0	5							0	4



0	1		0	1							2	0

Result : 20

PRACTICE SHEET – 2.5

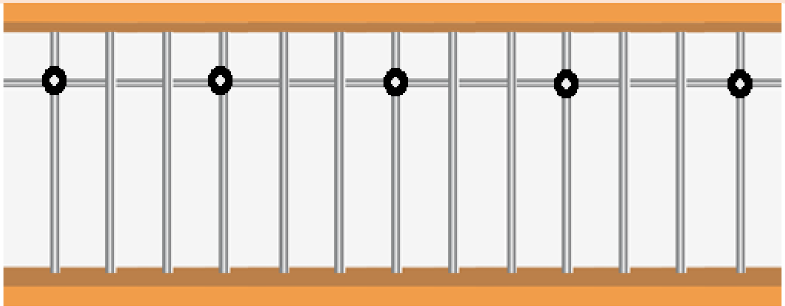
Solve the following:

a)	LCM (24 , 48)	=		e)	LCM (17 , 34)	=		i)	LCM (2 , 4 , 6)	=	
b)	LCM (13 , 39)	=		f)	LCM (95 , 76)	=		j)	LCM (5 , 10 , 15)	=	
c)	LCM (10 , 20)	=		g)	LCM (8 , 16)	=		k)	LCM (3 , 6 , 9)	=	
d)	LCM (11 , 33)	=		h)	LCM (6 , 12 , 18)	=		l)	LCM (15 , 30 , 45)	=	

PRACTICE SHEET – 2.6

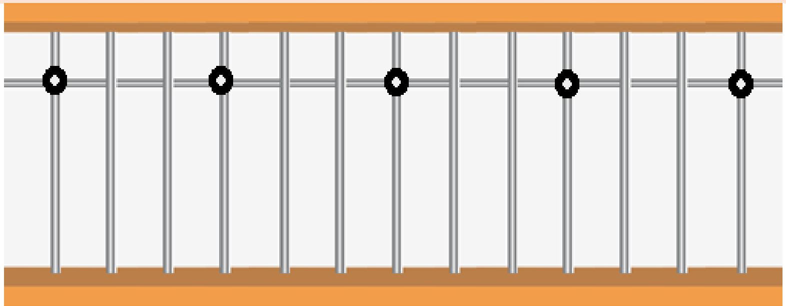
Solve the following using abacus representation to find prime factors:

a) LCM (12 , 16)



Result :

b) LCM (4 , 8 , 12)



Result :